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MS4092 COMPREHENSIVE EXAMINATION

Energy Conversion

Semester II 2022/2023

Saturday, 4th Mar 2023, Duration: 34 minutes

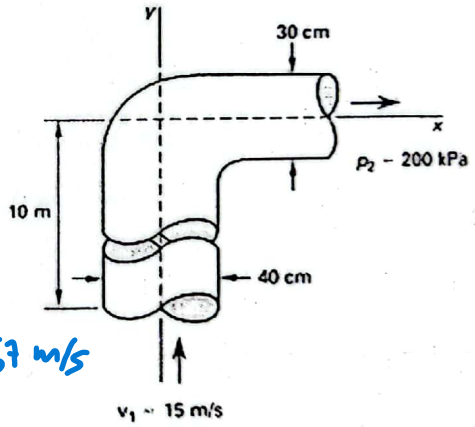
Closed Book and Closed Note

Answer all the questions on this problem sheet. State your answer by circling the letter in front of the most correct/accurate answer. Correct answers get additional scores, while wrong answers or questions that are not answered get zero score. If necessary, you may use the back page of the question sheet as 'scratch paper'. Don't forget to fill in your name, student identity number, and/or your signature at the top of EVERY page (in the space provided).

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1. The hydraulic radius of a fluid conduit is the
   a) mean radius from the center of flow to the wetted side of the conduit
   b) cross-sectional area of the conduit divided by the wetted perimeter
   c) wetted perimeter of the conduit divided by the area of flow
   d) cross-sectional area in flow divided by the wetted perimeter

2. A 90° reducing elbow connects x-and-y direction of a horizontal water piping shown in the given sketch. Water discharges to the right, as shown. How large is the x-direction force (Fx) needed to hold the reducer elbow in a stationary position? (Choose the closest option)



   a) 24 kN to the right
   b) 57 kN to the left
   c) 64 kN to the right
   d) 71 kN to the left

$A_1V_1 = A_2V_2$

$V_2 = \frac{15 \cdot 40^2}{30^2} = 26{,}67$ m/s

$P_2A_2 - F_x = -\dot{m}V_2$

$F_x = 2\times10^5 \cdot \frac{\pi}{4} \cdot 0{,}3^2 + 1000 \cdot \frac{\pi}{4} \cdot 0{,}3^2 \cdot 26{,}67^2$

$F_x = 64$ kN

3. The throttling process can be used to determine the vapor quality of substances that are in the liquid-gas mixture state based on the process principle:
   a) An isenthalpic pressure decreases to a saturated vapor state
   b) An isenthalpic pressure decreases to a saturated liquid state
   c) An isenthalpic pressure decreases to a superheated vapor state
   d) An isenthalpic pressure decreases to a subcooled liquid state

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$T_1 = 303\ K$ $T_2 = 288\ K$

4. A heat pump is used to maintain a hall at 30°C when the atmospheric temperature is 15°C. The heat loss from the hall is 1200 kJ/min. Calculate the power (kW) required to run the heat pump if its C.O.P. is 40% of the Carnot machine working between the same temperature limits.

a) 0.495 kW
b) 4.04 kW
c) 2.47 kW
d) 8.08 kW

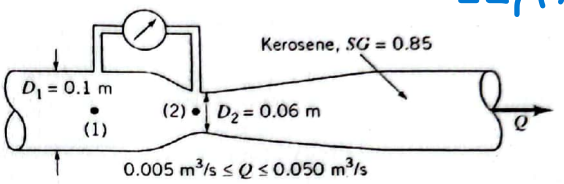
$Q_H = 1200\ \frac{kJ}{min} = \frac{1200}{60}\ kJ \left|\frac{1\ min}{60\ s}\right| \left|\frac{1\ kW}{1\ kJ}\right| = 20\ kW$

Ideal COP: $\frac{T_1}{T_1 - T_2} = \frac{303}{15} = 20.2$

Actual COP: $0.4 \times 20.2 = 8.08$

$\dot{W} = \frac{Q_H}{COP} = \frac{20}{8.08}\ kW = 2.47\ kW$

5. Venturimeter shown in the picture has kerosene flowing with a specific weight of SG=0.85, pipe diameter dimensions D_1=0.1 m, and throat diameter D_2=0.06 m. If the pressure difference is 116 kPa, what is the flow rate of Q?



a) 50 m^3/menit
b) 50 liter/s
c) 50 liter/menit
d) 50 m^3/s

$\Delta p = \frac{1}{2}\rho(V_2^2 - V_1^2)$ $Q = A_1V_1 = A_2V_2$

$\Delta p = \frac{1}{2}\rho\left(\frac{Q^2}{A_2^2} - \frac{Q^2}{A_1^2}\right)$ $Q = 0.05\ m^3/s$

$= 50\ l/s$

$Q^2 = \frac{2\Delta p}{\rho} \cdot \frac{1}{\left(\frac{1}{A_2^2} - \frac{1}{A_1^2}\right)}$

6. A bar with 1 x 1 cm cross section and length L = 20 cm lies horizontally between two vertical walls at the same temperature of 150°C. The rod conductivity k = 200 W/m.K. If there is convective heat transfer on the surface of the rod with a convection coefficient of h = 25 W/m^2.K and an ambient air temperature of T_∞ = 25°C, then the rate of heat transfer released by the rod is:

(a) 10,8 W (b) 15,7 W (c) 21,6 W (d) 31,4 W

Case	Tip Condition* ($x = L$)	Temperature Distribution θ/θ_b*	Fin Heat Transfer Rate q_f*
A	Infinite fin ($L \to \infty$): $\theta(L) = 0$	e^{-mx} $m = \sqrt{\frac{hP}{kA_c}}$	$M = \sqrt{hPkA_c}\,\theta_b$
B	Adiabatic: $d\theta/dx\|_{x=L} = 0$	$\frac{\cosh m(L-x)}{\cosh mL}$	$M \tanh mL$
C	Prescribed temperature: $\theta(L) = \theta_L$	$\frac{(\theta_L/\theta_b)\sinh mx + \sinh m(L-x)}{\sinh mL}$	$M\frac{(\cosh mL - \theta_L/\theta_b)}{\sinh mL}$
D	Active, convection heat transfer: $h\theta(L) = -kd\theta/dx\|_{x=L}$	$\frac{\cosh m(L-x) + (h/mk)\sinh m(L-x)}{\cosh mL + (h/mk)\sinh mL}$	$M\frac{\sinh mL + (h/mk)\cosh mL}{\cosh mL + (h/mk)\sinh mL}$

$\theta_b = T_b - T_\infty = 125\ K$ $m = 7.07$

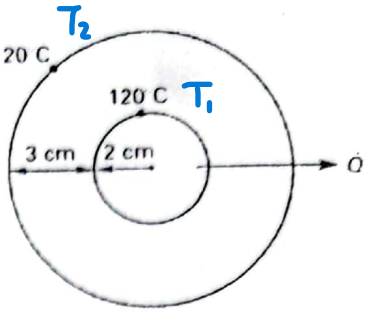
$A_c = w \times t = 1\ cm^2 = 0.0001\ m^2$ $q = 15.7\ W$

$P = 2w + 2t = 4\ cm = 0.04\ m$

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7. A 4 cm diameter steel pipe covered with 3 cm thick insulation carries steam at 120°C. The thermal conductivity of the insulation is 0.04 W/m·°C. If the resistance to heat transfer of the steel pipe is negligible, what is the thermal resistivity between the steam and the ambient air?



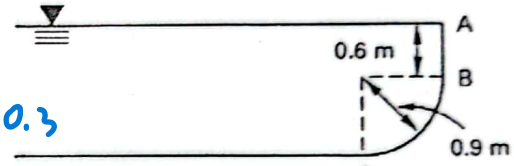
$k = 0,04$ W/m.°C

$r_i = 0,02$ m $\quad r_o = 0,05$ m

$R = \frac{\ln(r_o/r_i)}{2\pi k} = 3,645$ m.°C/W

a) 0.90 m.°C/W
b) 3.6 m.°C/W
c) 9.9 m.°C/W
d) 11 m.°C/W

8. The water tank shown has a width of 0.3 m. The rounded corner has a radius of 0.9 m. The magnitude of the water force over the length of wall from point A to point C is most nearly (use $g = 9.81$ m/s²)



$F_H = \gamma h_c A = 9,81 \times 1000 \times \frac{1,5}{2} \times 1,5 \times 0,3 = 3310,875$ N

$F_V = \gamma V = 9,81 \times 1000 \times (0,6 \times 0,9 + \frac{\pi}{4} \cdot 0,9^2) \times 0,3$

$F_V = 3461,48$ N $\quad F_R = \sqrt{F_V^2 + F_H^2} = 4800$ N

a) 3300 N
b) 3400 N
c) 4800 N
d) 7400 N

9. A solid at temperature T_s is introduced into a pool of fluid at temperature T_f. The solid temperature can be assumed to be uniform by taking into account the relevant dimensionality numbers, i.e. if:

a) Nu is large value
b) Ra is large value
c) Bi is small value
d) St is small value

10. Calculate the radiant flux density from a black surface at 400 degree Celsius?

$T = 400 + 273 = 673$ K

$Q = \varepsilon \sigma T^4 = 1 \times 5,67 \times 10^{-8} \times 673^4$

$Q = 11631,7$ W/m²

a) 1631.7 W/m²
b) 31.7 W/m²
c) 631.7 W/m²
d) 11631.7 W/m²

11. For a reversible heat transfer and process being adiabatic, which of the following is true?

$dQ = 0 \rightarrow$ reversible

$dS = 0 \rightarrow$ adiabatic

a) $dQ = 0, dS \neq 0$
b) $dQ = 0, dS = 0$
c) $dQ \neq 0, dS \neq 0$
d) $dQ \neq 0, dS = 0$

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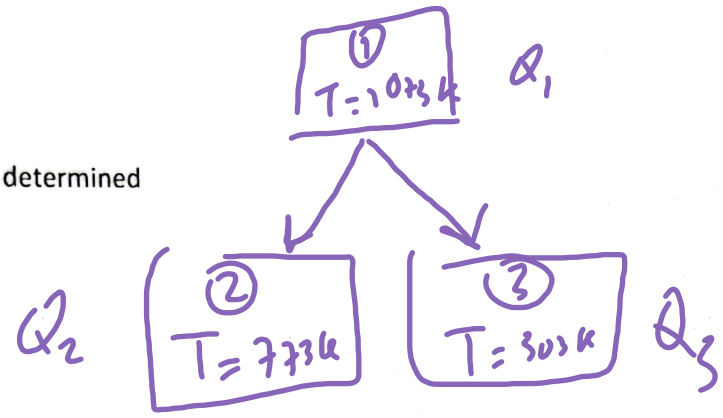
12. A reversible power generating engine "3TR" operates by taking heat from the thermal reservoir T_H = 800°C at a rate of 8 kW and dissipating heat to the thermal reservoir T_I = 500°C at a rate of 5 kW and to an environment T_C = 30°C at an unknown rate. The thermal efficiency of the power plant is

a) 61%

~~b)~~ 33.75%

c) 71.8%

d) Insufficient data, cannot be determined



1st law thermodynamics $Q_1 = Q_2 + Q_3 + W$

Clausius inequality $\frac{Q_1}{T_1} = \frac{Q_2}{T_2} + \frac{Q_3}{T_3}$

$\frac{8}{1073} = \frac{5}{773} + \frac{Q_3}{303}$

$Q_3 = 0,299\ kW$

$W = 2,701\ kW$

$\eta = \frac{W}{Q_H} = \frac{2,701\ kW}{8\ kW}$

$\eta = 0,3376$

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